

Ab-initio Molecular Dynamics: Beyond the Classical Picture

PWF Summer school on Theoretical and Computational Biophysics

University of Colombo

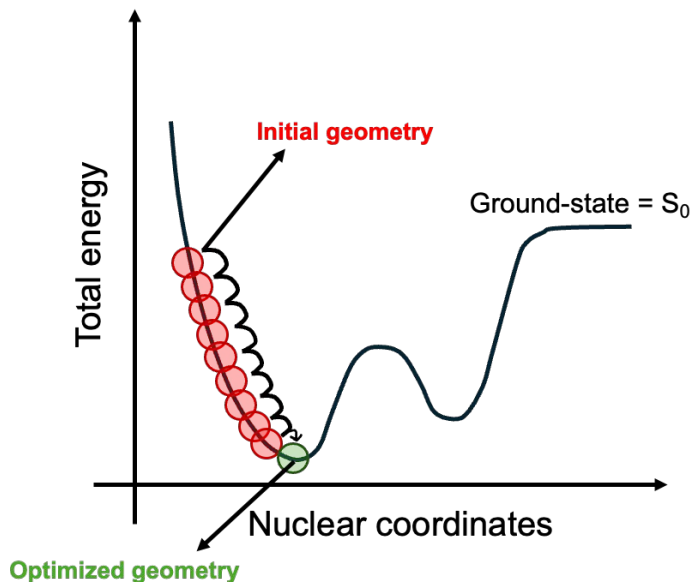
Day 4/5

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Static DFT

Static DFT (what we just saw):

- Geometry optimization → minimum-energy structure
- One geometry, ground-state energy and forces
- No thermal motion, no dynamics, no time



**Is this picture useful
for all applications?**

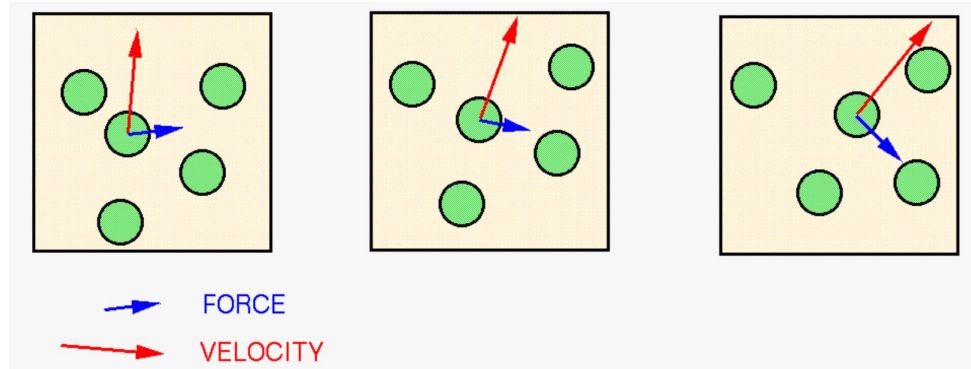
What is missing?

There are physical/chemical applications that require:

- **Finite temperature** → nuclei are never still: they vibrate, rotate, and diffuse. The bottom of the well is not where the system actually lives. Temperature also drives fluctuations that can access configurations far from the minimum
- **Time evolution** → bonds vibrate, molecules rearrange, reactions proceed. Static DFT gives one snapshot; we need a trajectory.

Can't we just use classical molecular dynamics for these applications?

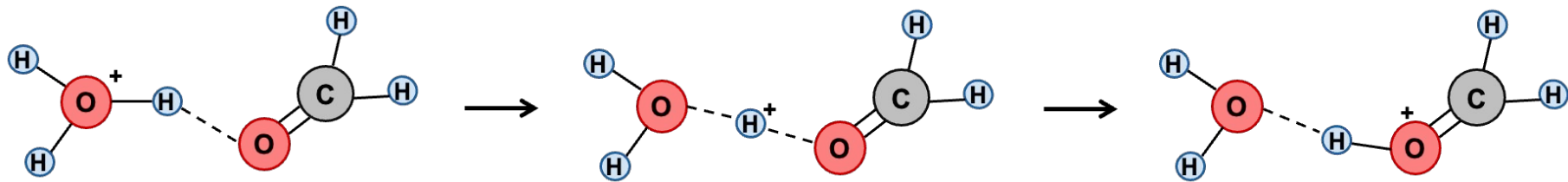
Limitations of classical force fields



but...

Molecular dynamics (MD) simulation
=
Forces + equation of motion $\rightarrow$ time evolution

Limitations of classical force fields



Classical MD wouldn't be able to describe/follow a reaction!

- Force fields can't describe changes in the electronic structure
- Force fields are fitted to specific bonding topologies → they don't transfer



Molecular dynamics + first-principles description of the electronic structure

DFT-based molecular dynamics

Electrons $\longrightarrow$ Atomic Forces $\longrightarrow$ Atomic Motion

$$H\Psi = E\Psi$$

$$F = m \times a$$

Electrons:

- Ground-state electronic structure computed for the instantaneous nuclear positions
- Provides the energy and forces on each atom

Atomic Motion:

- Temporal evolution of the atomic positions based on DFT forces
- Same MD machinery as before

Two main ab-initio MD flavors:

- Born-Oppenheimer Molecular Dynamics
- Car-Parrinello Molecular Dynamics

Born-Oppenheimer molecular dynamics

Why can we separate electronic and nuclear motions?

Born-Oppenheimer (BO) MD works within the **BO approximation** (see Lecture 10): electrons are much lighter and move much faster than nuclei → electrons are assumed to adjust instantaneously to the positions of the nuclei.

For nucleus I , with position $\mathbf{R}_I$:

$$M_I \ddot{\mathbf{R}}_I = F_I = -\nabla_I E_{BO}(\mathbf{R}) \quad \text{Equation of motion}$$

$$E_{BO}(\mathbf{R}) = \min_{\boldsymbol{\varphi}} E_{KS}(\boldsymbol{\varphi}, \mathbf{R}) + V_{NN}(\mathbf{R}) \quad \begin{array}{l} \text{BO potential} \\ \text{energy surface} \end{array}$$

$$\mathbf{R} = \{\mathbf{R}_1, \mathbf{R}_2, \dots, \mathbf{R}_{Nuc}\}$$

$$\boldsymbol{\varphi} = \{\varphi_i\}$$

Born-Oppenheimer molecular dynamics

For the nuclear positions $\mathbf{R}(t)$:

1. Solve the electronic problem $\rightarrow$ determine E_{KS} by direct minimization or self-consistent diagonalization of H_{KS}
2. Evaluate forces in the ground-state using the Hellmann-Feynman theorem (next slide)
3. Propagate in time (Newton's equations)
4. Repeat from step 1 for the new nuclear positions

- A new electronic-structure calculation is performed at every MD step $\rightarrow$ **high computational cost.**
- **No explicit electronic dynamics:** electrons are re-optimized to the instantaneous GS at each step.
- The electronic solution and forces must be sufficiently accurate

Hellmann-Feynman theorem

Once we have the electronic ground-state, how do we obtain the force on the nucleus I ?

$$F_I = -\nabla_I E_{BO}$$

We differentiate the energy:

$$\nabla_I E = \langle \nabla_I \Psi_0 | \hat{H} | \Psi_0 \rangle + \langle \Psi_0 | \nabla_I \hat{H} | \Psi_0 \rangle + \langle \Psi_0 | \hat{H} | \nabla_I \Psi_0 \rangle$$

and use:

$$\hat{H} | \Psi_0 \rangle = E_0 | \Psi_0 \rangle \quad \langle \Psi_0 | \Psi_0 \rangle = 1$$

Hellmann-Feynman theorem

$$\nabla_I E = \underbrace{\langle \nabla_I \Psi_0 | \hat{H} | \Psi_0 \rangle} + \underbrace{\langle \Psi_0 | \nabla_I \hat{H} | \Psi_0 \rangle} + \underbrace{\langle \Psi_0 | \hat{H} | \nabla_I \Psi_0 \rangle}$$

For an exact eigenstate and a complete basis set, the **contribution due to the variation of the wavefunction** cancels exactly.

$$F_I = -\langle \Psi_0 | \nabla_I \hat{H} | \Psi_0 \rangle$$

In this **exact ground-state limit**, we do not need to calculate explicitly how the wavefunction changes when the nuclei move.

BOMD in practice: cost and stability

BO molecular dynamics requires a converged electronic structure at every step.

Two consequences:

Cost:

- Electronic GS must be converged at every time step → dominant cost
- New nuclear geometry → new DFT calculation
- Timescales reachable in practice: ps to tens of ps

Is this timescale enough for the process we want to study?

Stability:

- HF assumes an exact ground-state
- Incomplete electronic convergence → **errors in the forces**
- Force errors can accumulate → **energy drift over time**
- Accurate BOMD therefore requires **tight electronic convergence**

Car-Parrinello molecular dynamics



Car-Parrinello (CP) molecular dynamics was developed in 1985 in Trieste (@ SISSA)

- Electronic degrees of freedom are introduced as **fictitious dynamical variables**, defining an extended **Lagrangian** that gives coupled equations of motion for ions and electrons.
- No need to re-optimize the electronic structure at each time step: the KS orbitals are propagated dynamically and, under adiabatic separation, remain close to the instantaneous electronic ground state, giving accurate forces on the ions.
- Avoids repeated SCF convergence at every step and eliminates SCF-convergence-induced force drift.

The Lagrangian formulation

A general framework for deriving equations of motion:

$$L(\mathbf{R}, \dot{\mathbf{R}}) = \sum_{I=1}^{Nuc} \frac{1}{2} M_I \dot{\mathbf{R}}_I^2 - V(\mathbf{R})$$

Lagrangian

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{\mathbf{R}}_I} - \frac{\partial L}{\partial \mathbf{R}_I} = 0$$

Euler-Lagrange equations

- The Lagrangian framework lets us derive equations of motion for any system for which we know the kinetic and potential energy.
- It is more general than Newton's formulation because the dynamical variables do not have to be only the positions of particles.

In CPMD, we use this flexibility to describe **a joint dynamics of the nuclear positions and the KS orbitals.**

The Car-Parrinello Lagrangian

$$L(\mathbf{R}, \dot{\mathbf{R}}, \boldsymbol{\varphi}, \dot{\boldsymbol{\varphi}}) = \underbrace{\sum_I \frac{1}{2} M_I \dot{\mathbf{R}}_I^2}_{\text{Nuclear kinetic}} + \underbrace{\mu \sum_i \langle \dot{\varphi}_i | \dot{\varphi}_i \rangle}_{\text{Fictitious electronic kinetic}} - \underbrace{E[\boldsymbol{\varphi}, \mathbf{R}]}_{\text{Potential energy}} + \underbrace{\sum_{i,j} \Lambda_{ij} (\langle \varphi_i | \varphi_j \rangle - \delta_{ij})}_{\text{orthonormality constraint}}$$

- μ = fictitious mass parameter $\rightarrow$ gives inertia to the KS orbitals
- Nuclear positions and KS orbitals are now propagated simultaneously

Coupled equations of motion:

$$\begin{aligned} \frac{d}{dt} \frac{\partial L_{CP}}{\partial \dot{\mathbf{R}}_I} &= \frac{\partial L_{CP}}{\partial \mathbf{R}_I} & \frac{d}{dt} \frac{\delta L_{CP}}{\delta \langle \dot{\varphi}_i |} &= \frac{\delta L_{CP}}{\delta \langle \varphi_i |} \\ M_I \ddot{\mathbf{R}}_I &= -\frac{\partial E}{\partial \mathbf{R}_I} + \sum_{ij} \Lambda_{ij} \frac{\partial}{\partial \mathbf{R}_I} \langle \varphi_i | \varphi_j \rangle & \mu |\ddot{\varphi}_i\rangle &= -\frac{\delta E}{\delta \langle \varphi_i |} + \sum_j \Lambda_{ij} |\varphi_j\rangle \end{aligned}$$

Adiabatic separation in CPMD

How do we remain close to the BO surface?

- Fictitious electronic motion must be much faster than nuclear motion:

$$\omega_{el} \gg \omega_{nuc} \quad \omega_{el} \propto \frac{1}{\sqrt{\mu}}$$

- Electronic and nuclear frequencies are well separated $\rightarrow$ little transfer of energy between the two sets of degrees of freedom $\rightarrow$ the KS orbitals remain close to the instantaneous GS (oscillations around the BO minimum) $\rightarrow$ cold electrons
- If μ is too large $\rightarrow$ nuclear/electronic coupling (no adiabatic separation)
- If μ is too small $\rightarrow$ electronic frequencies are too high $\rightarrow$ we need a much smaller Δt

BO vs CP molecular dynamics

Born-Oppenheimer MD

- Exactly on BO surface; in principle more accurate
- $\Delta t \approx$ ionic time scales (maximum time step possible)
- Expensive minimization at each MD step
- Energy drift possible due to incomplete SCF convergence (non-conservative forces)

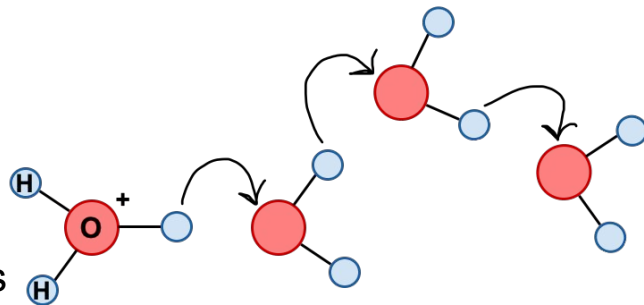
Car-Parrinello MD

- Always slightly off BO surface $\rightarrow$ orbitals oscillate around the minimum
- $\Delta t \ll$ ionic time scales (much shorter time step necessary)
- No SCF minimization at each step $\rightarrow$ cheaper per MD step
- Stable dynamics: no systematic energy drift (in the adiabatic regime)

Application: proton transfer in aqueous systems

Why proton transfer (PT) matters:

- Ubiquitous elementary step in chemistry and biology
- Acid–base chemistry, pH, autoionization of water
- Enzymatic catalysis (e.g. proton relays in active sites)
- Electrochemistry, fuel cells, proton-conducting membranes



Grotthuss mechanism → proton “hops” along the H-bond network → **AIMD can capture bonds breaking and forming**

However, we have a timescale issue: individual PT events are fast, but rare on the AIMD timescale (tens of ps) → poor statistics on PT rates, mechanisms, etc... Enhanced sampling helps, but is expensive



Machine learning interatomic potentials